

Effectiveness of Marketing Campaigns by PopCrt from a Demographic Perspective

Problem

PopCrt, a fast-growing online shopping company from the Philippines specializing in lifestyle and fashion brands from independent stores across the country, wants to know if their marketing campaigns are effective in increasing clicks and profits, depending on the target market. The business objective is to determine which product categories are the most popular per market segment (e.g. gender, location, and age group), which campaigns are performing best by gender, and which age groups are more likely to interact with the marketing campaigns. The analysis goals are therefore as follows:

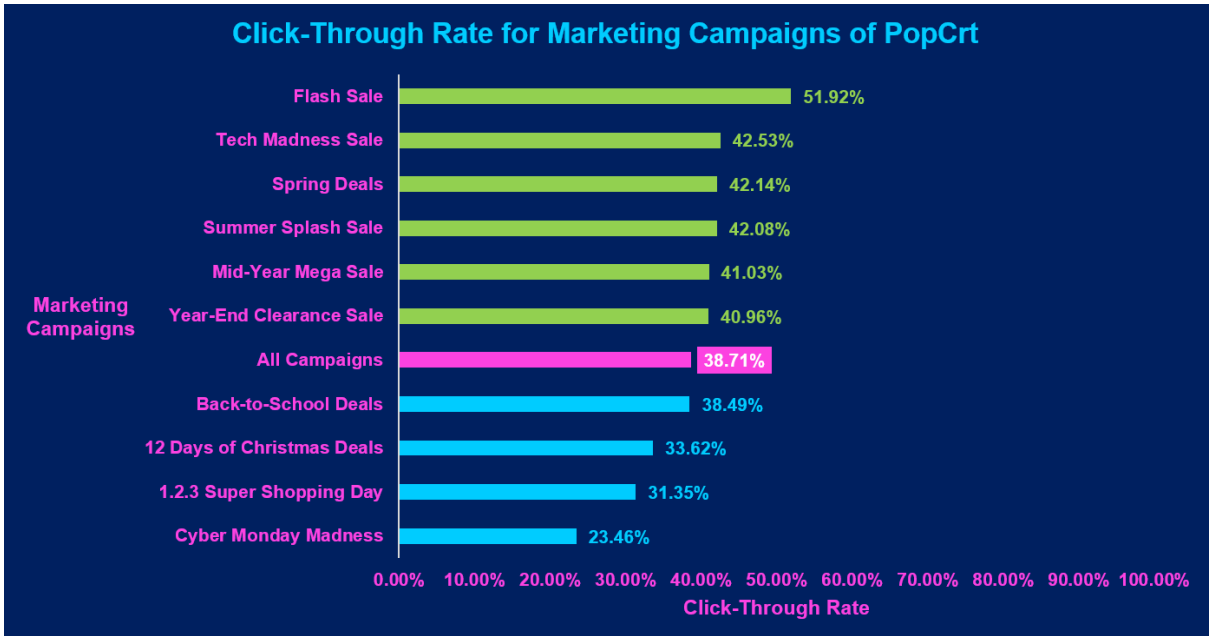
- What is the overall click-through rate (CTR) for the campaigns launched by the company?
- What is the CTR for each product category?
- What percentage of the total revenue generated during campaign periods is contributed by female customers?
- Which campaign has generated the most impressions?
- How many unique users per location viewed the campaign with the most impressions?
- What are the total impressions and CTR for each product category per location?
- What are the total impressions and CTR for each product category per age group?
- What are the total impressions and CTR for each product category per gender?

Methodology

PostgreSQL was used for cleaning and preprocessing the data. The sales data was first left-joined with the marketing campaign impressions data. The Click-Through Rate (CTR), defined as the total number of clicks divided by the total number of impressions (i.e. the number of times a marketing campaign was shown to customers whether clicked or not, calculated by counting the number of instances of a marketing campaign in the impressions dataset) and presented in percentage form, was calculated for each of the marketing campaigns and product categories, and as well throughout all marketing campaigns or product categories. The percentage of the total revenue generated during campaign periods was also calculated for female customers, overall and per marketing campaign. The total number of unique users per region was also calculated for the marketing campaign with the most number of impressions. Lastly, the total impressions and CTR of each product category was calculated per location, age group, and gender. The results are then transferred to an Excel spreadsheet for visualization.

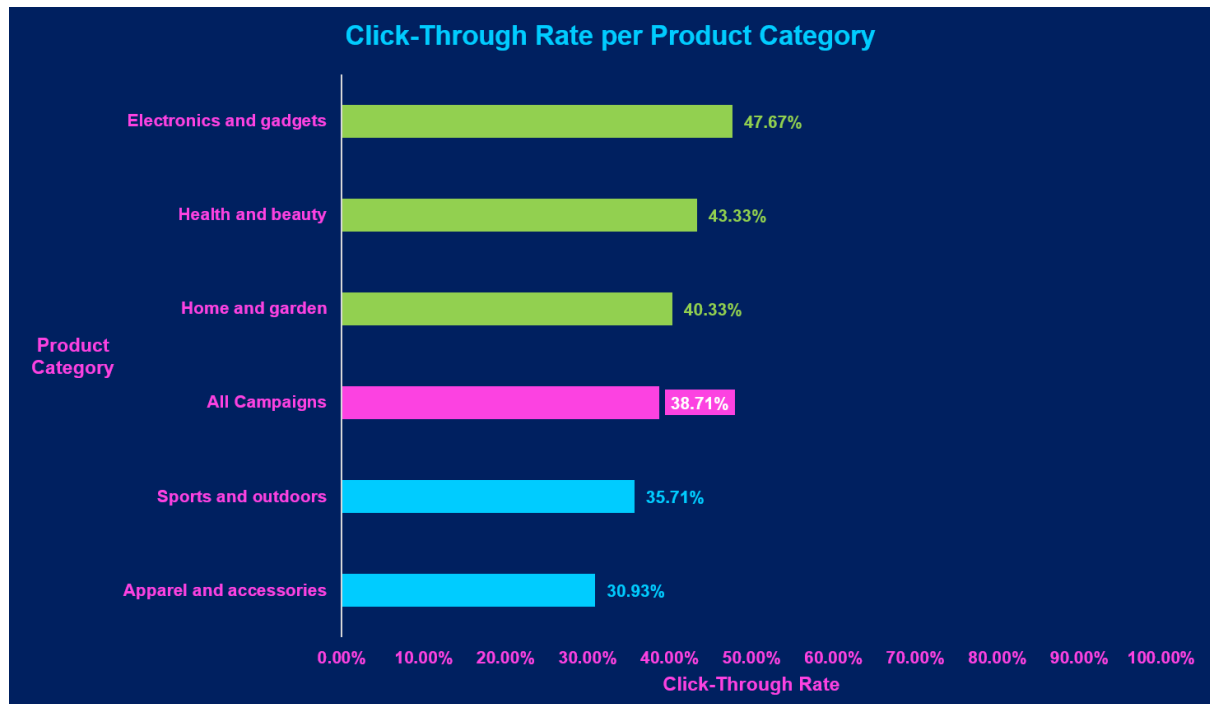
Findings

What is the overall click-through rate (CTR) for the campaigns launched by the company?



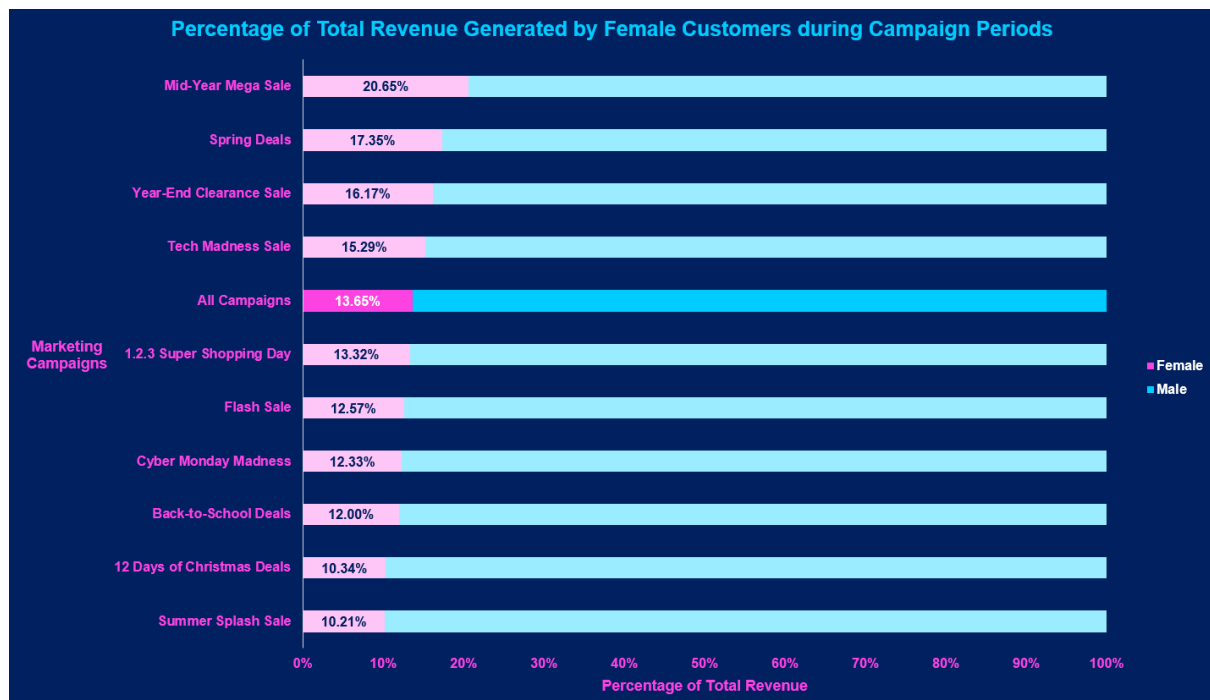
The click-through rate (CTR) for all the campaigns launched by the company is 38.71%, with Flash Sales generating the highest click-through rate of all the campaigns. On the other hand, Back-to-School Deals, 12 Days of Christmas Deals, 1.2.3 Super Shopping Day, and Cyber Monday Madness (which also has the lowest click-through rate of all the campaigns) are below the overall CTR. With the exception of Flash Sales, the click-through rate of the campaigns are below 50%.

What is the CTR for each product category?



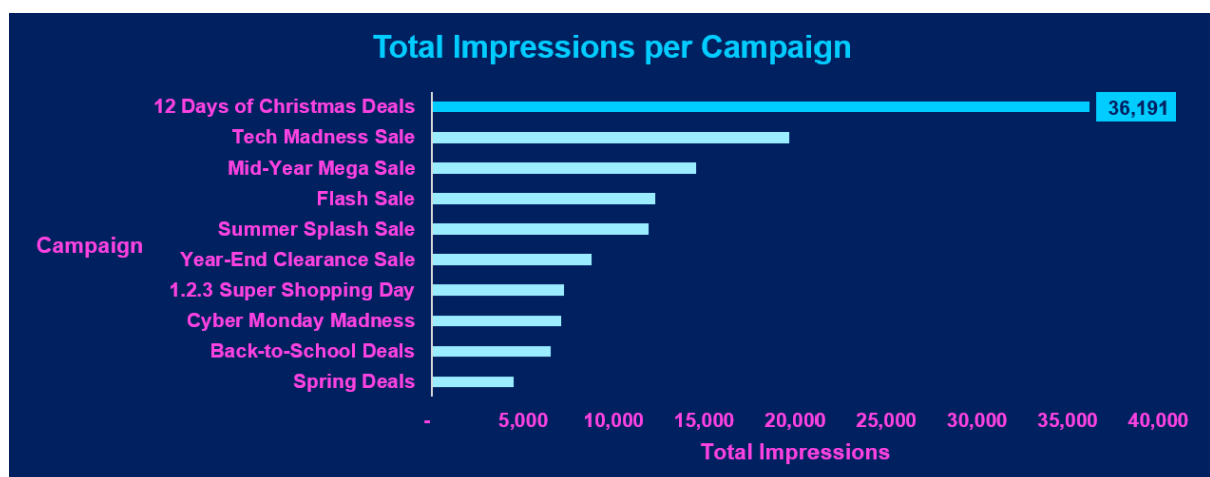
Of the product categories, Electronics and Gadgets had the highest click-through rate with a CTR of 47.67%, followed by Health and Beauty with 43.33%, Home and Garden with 40.33%, Sports and Outdoors with 35.71%, and lastly Apparel and Accessories with 30.93%.

What percentage of the total revenue generated during campaign periods is contributed by female customers?



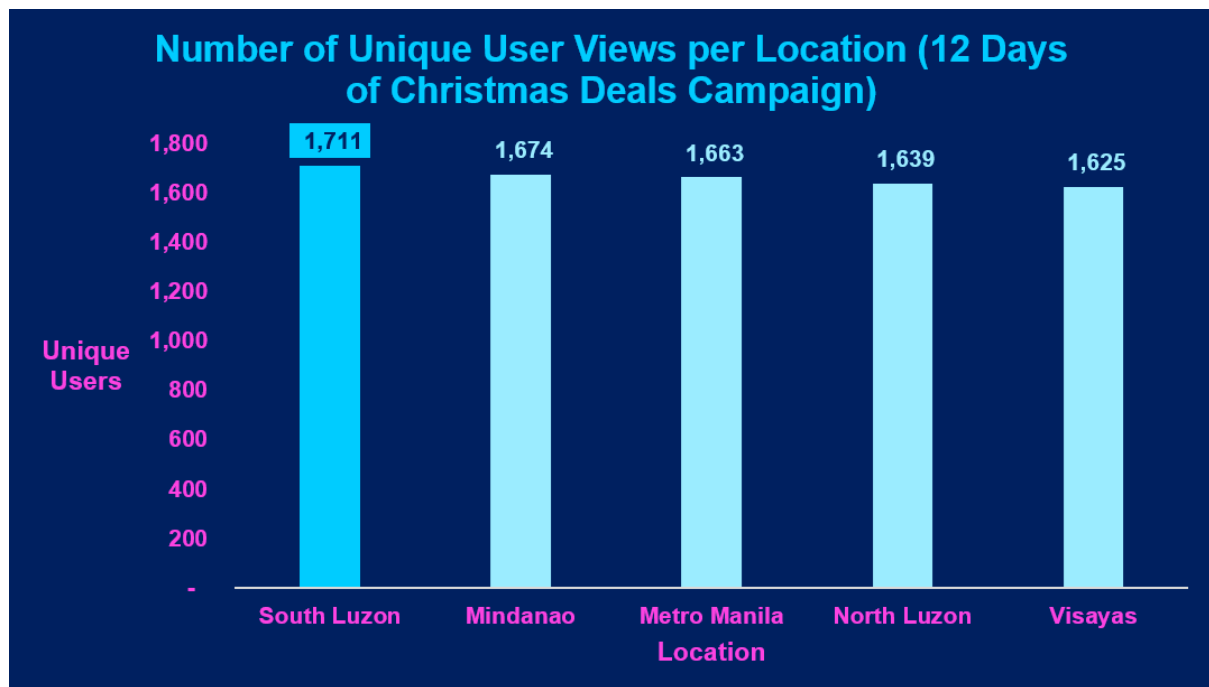
For all campaigns overall, 13.65% of the total revenue during campaign periods are generated by female customers. In addition, for any given marketing campaign, female customers only contribute less than 25% to the total revenue, with Mid-Year Mega Sale garnering the highest percentage of the total revenue generated by female customers with 20.65%.

Which campaign has generated the most impressions?



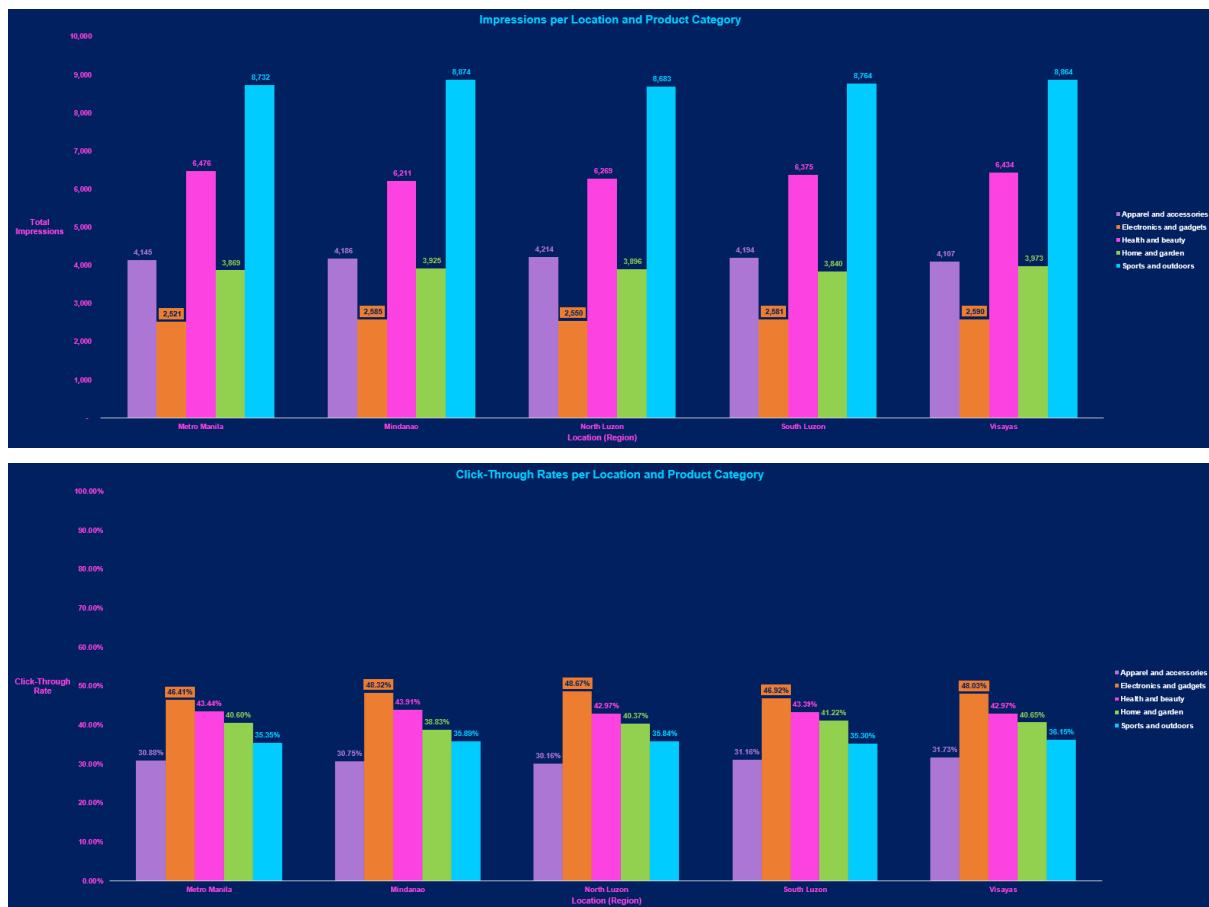
Of the campaigns, the 12 Days of Christmas Deals, which is one of the marketing campaigns with a CTR less than the overall, had the highest total impressions.

How many unique users per location viewed the campaign with the most impressions?



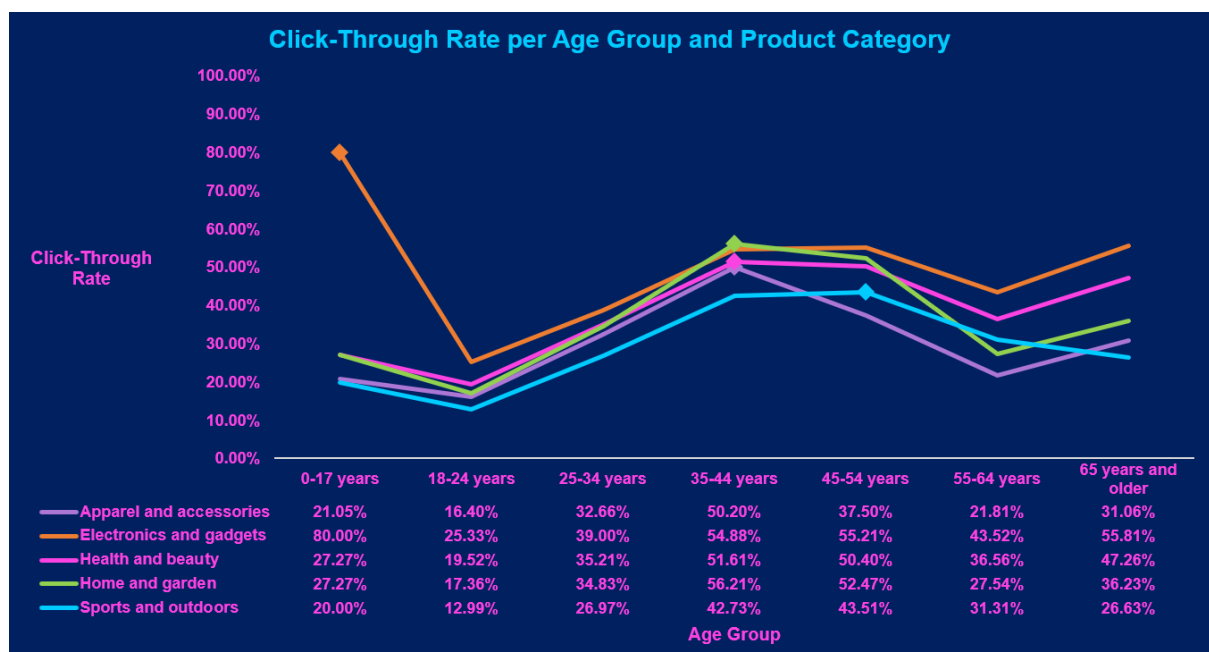
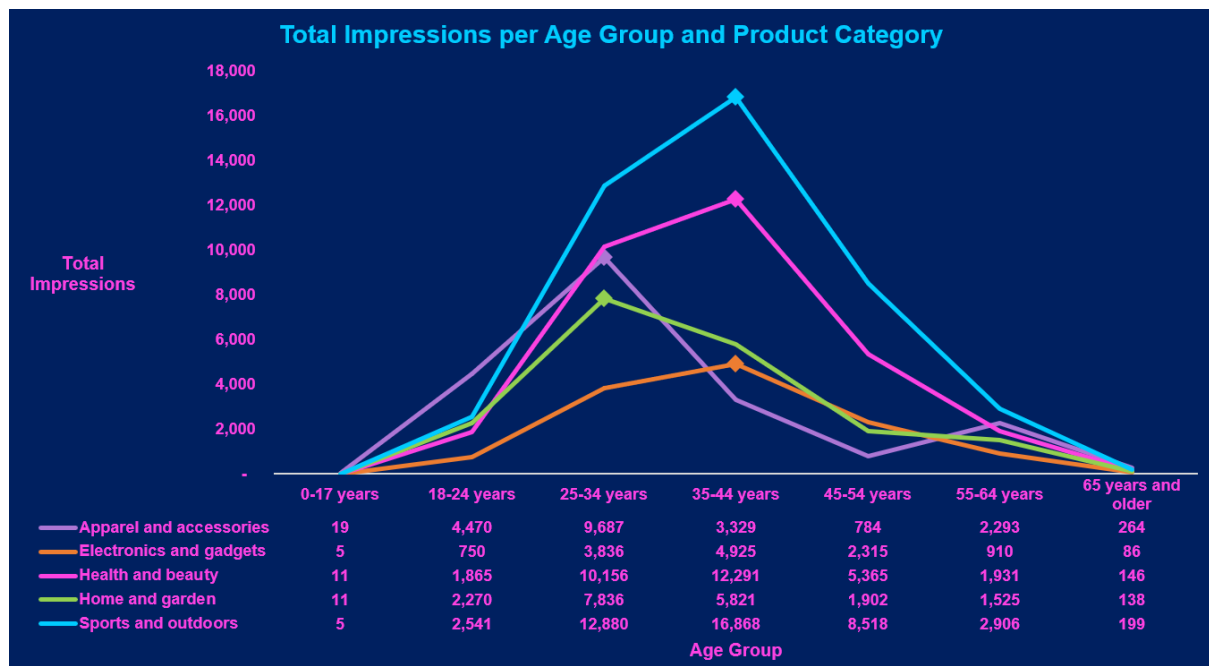
South Luzon had the highest number of unique users who viewed the campaign with 1,711 unique users. It is then followed by Mindanao with 1,674 unique users, Metro Manila with 1,663 unique users, North Luzon with 1,639 unique users, and lastly by Visayas with 1,625 unique users. Recalling that total impressions mean the number of times a content is displayed to users whether it got clicked or not, this fact on the number of unique views per location for the campaign with the highest impressions (i.e. 12 Days of Christmas Deals Campaign) could give insights on the maximum reach of the PopCrt campaign efforts nationwide. The reach is almost consistent for all the regions.

What are the total impressions and CTR for each product category per location?



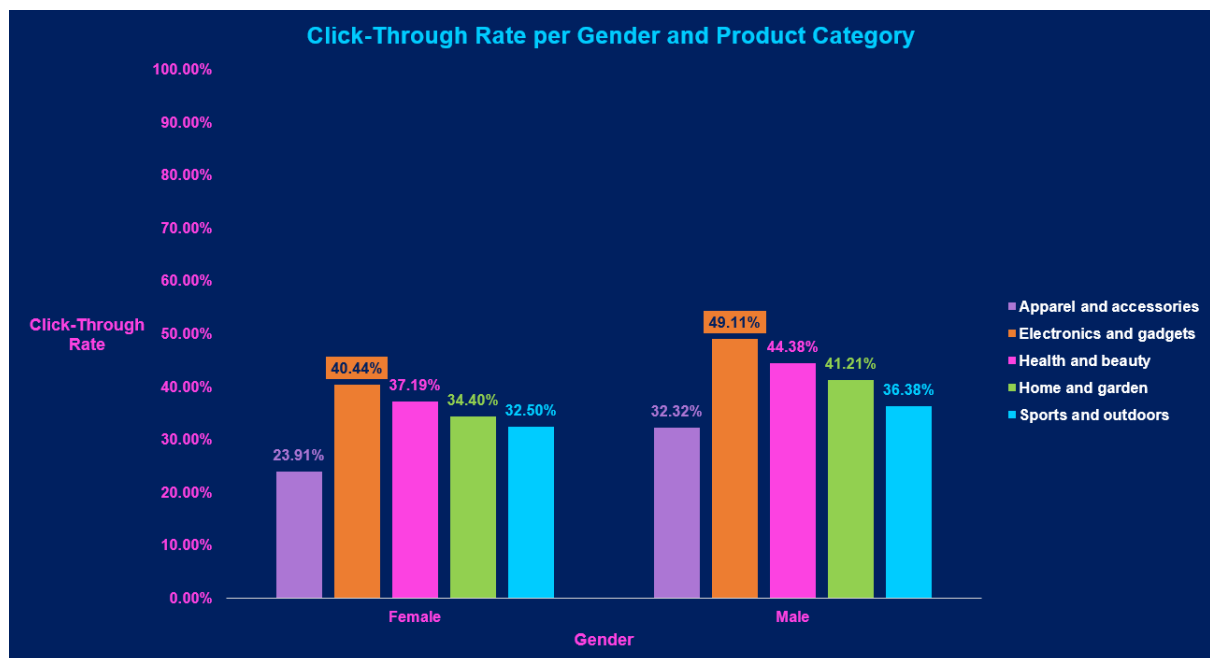
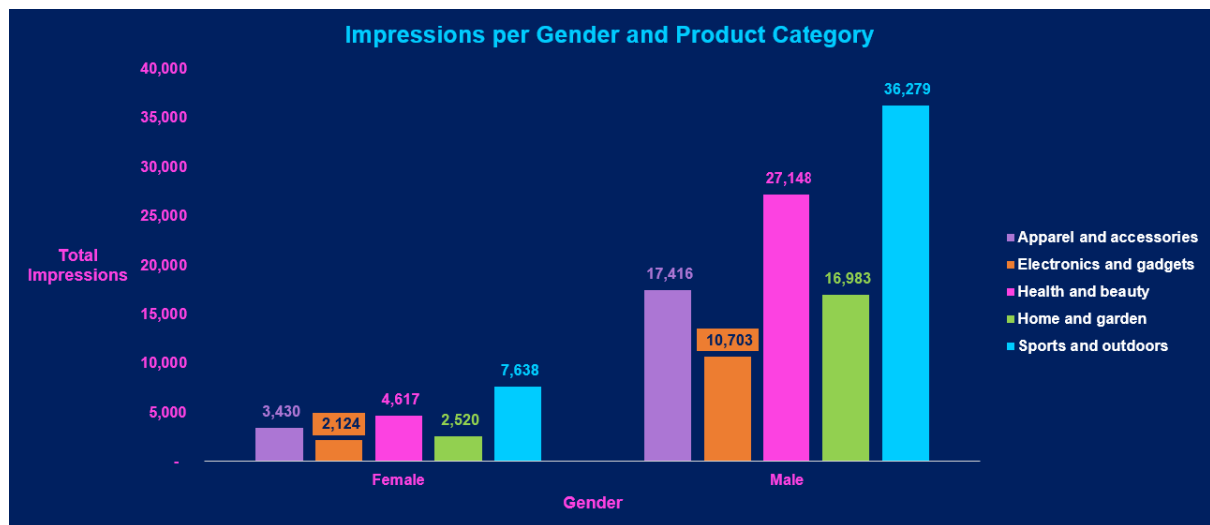
Across all locations, Electronics and Gadgets has the lowest impressions (Sports and Outdoors had the highest), but it has the highest click-through rate of all the product categories. The total number of impressions and the CTR values for each category are almost consistent across all locations.

What are the total impressions and CTR for each product category per age group?



Total Impressions peaked for Apparel and Accessories and Home and Garden in the age group 25-34 years, and in the age group 35-44 years for Health and Beauty, Sports and Outdoors, and Electronics and Gadgets. On the other hand, Click-Through Rate for Electronics and Gadgets peaked at 0-17 years (the age group with the lowest impressions for all product categories), 45-54 years for Sports and Outdoors, and at 35-44 years for all other product categories.

What are the total impressions and CTR for each product category per gender?



For both male and female customers, Electronics and Gadgets had the lowest total impressions but had the highest Click-Through Rate. When looking at the total impressions visual, there were significantly less campaigns presented to female users than male users, despite the CTR being almost consistent for both genders, which may help explain the low percentage of contribution to the total revenue by female customers.

Recommendations

Based on the findings, the recommendations for improving the marketing campaigns of PopCrt include the following:

- Focus more effort on marketing campaigns involving the Electronics and Gadgets product category, most especially in the 0-17 age group.
- Increase marketing campaign efforts on female users to potentially increase their chances of purchasing from the store.
- Adopt the marketing strategies utilized in Flash Sales to improve the CTR of marketing campaigns below the CTR of all campaigns, most notably the 12 Days of Christmas Deals which had the highest impressions but has one of the below-overall CTR values.